Teacher answers and topic guidance

Accept scientifically equivalent answers in the learner's usual mode. Award each distinct point once. Keep answers hidden during independent attempts.

Starter

1. Active transport.
2. Water.
3. The nucleus.

Full and faded sequence

Interphase; prophase; metaphase; anaphase; telophase; cytokinesis. The faded step is anaphase: the two copies of each chromosome are pulled to opposite ends. DNA is copied before mitosis, in interphase.

Spot the mitosis error

Fault 1: the first error is in interphase. Copy DNA before mitosis so both new nuclei can receive matching sets. Fault 2: the first error is in anaphase. Send one A copy to each end, with a B copy at each end. The repair view shows the resulting matching sets after the later final split. Switching cases resets the previous diagnosis and repair.

Six-chromosome hinge

C: six in each daughter, with the same set as the original body cell. A, three: confuses this ordinary body-cell model with chromosome halving. B, twelve: treats copied material as the number retained by each daughter. D, none: leaves the copies in one cell instead of distributing them.

About three in four secure responses: move on. Otherwise re-model with two blue and two orange paper strips, labelled A and B, then use the parallel case. A starting set of eight gives eight in each daughter.

Transfer to repair

DNA is copied before matching copies separate into two nuclei. After the cell splits, matching new body cells can replace damaged cells. Accept the same explanation through a model demonstration.

Science scope and misconceptions

Mitosis separates copied chromosomes into two nuclei. Cytokinesis splits the cell. The model concerns ordinary body-cell division. It does not teach the full molecular detail, asexual reproduction or chromosome halving. Do not describe mitosis alone as the whole final split.

Copy before separation: both new nuclei need a matching set. Uneven separation can leave one nucleus with an extra copy and another lacking it. New matching body cells support growth and replacement of damaged cells.

Arrival Supported

1. Two cells.
2. The one cell on the left.
3. The inner outline labelled nucleus.
4. One cell has become two; each new cell has a matching labelled set.

Arrival Independent

1. Two.
2. The nucleus.
3. Both cells have A and B: the sets match.
4. One cell became two, each with a matching chromosome set.

Arrival Aspire

1. Two separate cells.
2. The nucleus.
3. Each new cell has both types, A and B, so the sets match.
4. It shows one cell becoming two with matching sets; it does not show the full molecular detail or every real chromosome.

Exit Supported

1. Before mitosis, in interphase.
2. Anaphase.

Exit Independent

1. Anaphase, when copies separate.
2. Send one A copy and one B copy to each end so each new nucleus receives both types.

Exit Aspire

1. Repair copying in interphase first. The copies must exist before the later separation can work.
2. Splitting the cell does not redistribute chromosome copies. Repair their separation into matching sets before the final split.

Made by Matt

Mark scheme Supported

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Order interphase, mitosis and cytokinesis. Show A and B in each new cell. [3]

Interphase, mitosis, cytokinesis; matching A and B sets in both new cells. Award one point for each correctly placed stage, with matching sets checked as the success criterion.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

2. Complete the faded sequence: prophase, metaphase, ___, telophase. Say what the missing step does. [2]

Anaphase; the two copies of each chromosome move to opposite ends.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

3. Fault 1 skips DNA copying. Choose interphase or cytokinesis as the first step to repair. Explain why. [3]

Interphase; copy DNA before mitosis; both new nuclei need a matching set of copies.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt

Mark scheme Independent

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Order interphase, prophase, metaphase, anaphase, telophase and cytokinesis. Mark the four phases that belong to mitosis. [4]

Interphase first; prophase then metaphase; anaphase then telophase; cytokinesis last and the four middle phases identified as mitosis.

Maximum 4 marks. Accept equivalent valid wording or a clear model demonstration.

2. Fault 1 goes into mitosis without copying DNA. Find the first wrong step and explain its repair. [3]

First error in interphase; copy DNA before mitosis; this supplies matching copies for both new nuclei.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

3. Fault 2 sends both A copies to one new nucleus. Find the first wrong step, repair it and explain the result. [3]

First error in anaphase; send one A copy to each end; each new nucleus then receives a matching A and B set.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt

Mark scheme Aspire

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Fault finder: compare two faulty models. One skips DNA copying; the other puts both copies of one chromosome in one nucleus. Find each first error and propose a repair. [4]

Identify copying skipped in interphase; add the copying step before mitosis; identify both copies sent to one nucleus during separation; send one copy of each chromosome to each new nucleus before the final split.

Maximum 4 marks. Accept equivalent valid wording or a clear model demonstration.

2. Explain why swapping mitosis and DNA copying would break the model. [3]

DNA copying belongs before mitosis; without copies there is not a copied set to distribute to each new nucleus; both new cells would not receive the intended matching set.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

3. Predict what would happen to the sets if copied chromosomes did not separate evenly. Give a reason. [3]

New nuclei could have unequal sets; one could gain a chromosome copy and another lack it; the sets would no longer match because separation was uneven.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt

Mark scheme Choice

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Choose the first, middle and last stage from interphase, mitosis and cytokinesis. Show both A and B in each new cell. [3]

Interphase first, mitosis middle, cytokinesis last, with matching labelled sets checked.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

2. Choose anaphase or telophase for the missing step after metaphase. Show what happens to the copies. [2]

Anaphase; the two copies of each chromosome move to opposite ends.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

3. Fault 1 skips DNA copying. Choose copy DNA first or split the cell first as the repair. Give a reason. [3]

Copy DNA first; copying belongs in interphase before mitosis; each new nucleus needs a matching set.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt